SOFTWARE REQUIREMENTS SPECIFICATION

Human Resources Management System (HRMS)

Project Name	Aru HRMS
Document Type	Software Requirements Specification (SRS)
Version	1.0
Prepared For	College Mini Project
Platform	Web Application
Primary Technology	HTML, CSS, JavaScript, Node.js, MySQL

Confidentiality Notice: This document is prepared for academic/project use and describes the functional and non-functional requirements of the proposed HRMS.

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1. Introduction

1.1 Purpose

The purpose of the Human Resources Management System (HRMS) is to provide a centralized web-based platform for managing employee information and common HR activities. The system reduces manual record keeping, improves accessibility of employee data, and provides a structured dashboard for HR staff and employees.

1.2 Scope

The system covers user registration and login, employee management, employee profiles, department information, notifications, dashboard statistics, role-based access, and basic administrative functions. The initial design is suitable for approximately 100 employees and can be extended as the organization grows.

1.3 Intended Users

• HR/Admin: manages employees, roles, departments and notifications.
• Employee: views and updates permitted profile information and receives notifications.
• System Administrator: maintains system configuration and user access.

1.4 Definitions

HRMS – Human Resources Management System. **Admin** – authorized user who manages HR data. **Employee** – registered organization member. **RBAC** – Role-Based Access Control.

2. Overall Description

2.1 Product Perspective

Aru HRMS is a web application consisting of a browser-based frontend, a backend/API layer, and a relational database. The frontend provides the user interface, the backend validates requests and applies business rules, and MySQL stores persistent data.

2.2 Product Functions

• Secure login and registration
• Dashboard with HR/employee statistics
• Add, view, edit and manage employee records
• Employee profile management
• Department and role information
• Notifications for all employees or selected departments
• Theme preference (light/dark)
• Search and filtering of employee records
• Role-based access control
• Logout and session management

2.3 User Characteristics

Users are expected to have basic web application skills. HR/Admin users require additional permissions for employee and notification management.

2.4 Operating Environment

Client: modern Chrome, Edge, Firefox or equivalent browser. Server: Node.js runtime. Database: MySQL 8.x. Development environment: Windows with VS Code. The application should support desktop and responsive browser views.

3. System Features and Functional Requirements

FR-01: User Registration

The system shall allow a new user to register using required identity and account details. It shall validate mandatory fields and prevent duplicate accounts.

FR-02: Login and Authentication

The system shall authenticate users using valid credentials. Invalid credentials shall display an appropriate error message. The system shall provide logout functionality and protect authenticated pages.

FR-03: Dashboard

The system shall display a dashboard containing relevant summary information such as total employees, departments, notifications and recent activity. Dashboard content shall vary according to the user's role where applicable.

FR-04: Employee Management

Authorized HR/Admin users shall be able to add, view, edit and deactivate employee records. The system shall store employee ID, name, contact information, department, role/designation, joining date and other required profile fields.

FR-05: Employee Profile

Users shall be able to view their profile. Employees shall only edit fields permitted by their role, while authorized administrators may update employee records.

FR-06: Department and Role Management

The system shall associate employees with departments and roles. Authorized users shall be able to maintain these associations.

FR-07: Notifications

Authorized users shall be able to create notifications targeted to all employees or a specific department. Users shall be able to view unread and read notifications.

FR-08: Search and Filtering

Authorized users shall be able to search employees by relevant fields and filter records by department, role or status.

FR-09: Role-Based Access Control

The system shall restrict protected operations according to the user's role. Unauthorized users shall not be allowed to access administrative functions.

FR-10: Theme Preference

The interface shall support light and dark theme preferences. The selected preference shall be retained for the user where supported.

FR-11: Validation and Error Handling

The system shall validate input data on the client and server sides where appropriate, show clear validation messages, and handle failed requests without exposing sensitive system information.

4. External Interface Requirements

4.1 User Interface

The UI shall contain a login/registration area, sidebar or navigation menu, dashboard cards, employee tables, profile forms, notification area and theme control. The interface shall use consistent typography, spacing, buttons, forms and status indicators.

4.2 Software Interfaces

Frontend communicates with the Node.js backend through HTTP/REST-style API endpoints. The backend communicates with MySQL using a suitable database driver/ORM.

4.3 Hardware Interfaces

No specialized hardware is required. The system shall run on standard computers or mobile-capable browsers with network connectivity.

4.4 Communication Interfaces

The application shall use HTTP/HTTPS for client-server communication. HTTPS is recommended for deployment to protect credentials and other sensitive data.

5. Non-Functional Requirements

NFR-01 Security

Passwords shall not be stored as plain text. Authentication and authorization shall be enforced on protected API operations. User input shall be validated and database queries shall use safe parameterization.

NFR-02 Performance

Common operations such as login, dashboard loading and employee search should respond promptly under the expected initial workload of approximately 100 employees.

NFR-03 Usability

The application shall provide clear navigation, readable labels, understandable validation messages and consistent UI behavior.

NFR-04 Reliability

The system shall handle invalid requests gracefully and maintain data consistency during create/update operations.

NFR-05 Maintainability

Frontend, backend and database responsibilities should be separated. Code should use meaningful names and reusable modules/components.

NFR-06 Scalability

The database design and API structure shall permit the employee count and feature set to grow beyond the initial 100-employee requirement.

NFR-07 Compatibility

The system should work on current major browsers and common desktop screen sizes, with responsive behavior where practical.

6. Data Requirements

6.1 Main Entities

Entity	Important Attributes
User	user_id, employee_id, email, password_hash, role, status
Employee	employee_id, name, DOB, gender, phone, email, address, joining_date, department_id, role_id
Department	department_id, department_name, description
Role	role_id, role_name, permissions

Entity	Important Attributes
Notification	notification_id, title, message, target_type, department_id, created_by, created_at

6.2 Data Integrity

Employee IDs and user accounts shall be unique. Required fields shall not be empty. Foreign-key relationships should preserve valid department and role associations. Deactivation should be preferred over destructive deletion where historical records need to be retained.

7. Use Cases

UC-01 Login — Actor: User. User enters credentials → system validates → successful user is taken to the appropriate dashboard; otherwise an error is shown.

UC-02 Register — Actor: New User. User enters registration details → system validates → account is created if data is valid and unique.

UC-03 Add Employee — Actor: HR/Admin. Admin opens employee form → enters details → system validates → employee record is stored.

UC-04 Manage Employee — Actor: HR/Admin. Admin searches an employee → views/edits permitted fields → system validates and updates the record.

UC-05 View Profile — Actor: Employee/Admin. User opens profile → system retrieves permitted information → profile is displayed.

UC-06 Send Notification — Actor: HR/Admin. Admin creates message → chooses all employees or department → system publishes notification to matching users.

UC-07 Search Employee — Actor: HR/Admin. User enters search/filter criteria → system returns matching employee records.

UC-08 Logout — Actor: Authenticated User. User selects logout → session/token is invalidated or cleared → user returns to login page.

8. System Constraints and Assumptions

• The first release is intended for an initial organization size of about 100 employees, with extensibility for more.
• Users require valid accounts and appropriate permissions.
• The system depends on a working database and server environment.
• Email/SMS integration, payroll, biometric attendance and advanced recruitment workflows are outside the initial scope unless added in a future release.
• Production deployment should use HTTPS, secure secrets management, regular database backups and appropriate access controls.

9. Acceptance Criteria

The system will be considered acceptable when the following are demonstrated:
1. A valid user can register/login and logout successfully.
2. Unauthorized users cannot perform protected HR/Admin actions.
3. Authorized HR/Admin users can add, view, search and update employee records.
4. Employee profiles display the correct stored information.
5. Notifications can be targeted to all employees or a department and are visible to intended recipients.
6. Input validation prevents incomplete or invalid required data from being stored.
7. The application provides clear error/success feedback.
8. Core data remains consistent after normal create/update operations.

10. Future Enhancements

Possible future modules include attendance and leave management, payroll integration, recruitment/Applicant Tracking System, performance appraisal, document management, email notifications, analytics and reports, audit logs, employee self-service workflows, mobile/PWA support and advanced HR dashboards.

11. Document Approval

Role	Name / Signature	Date
Project Guide		
Student	Kowshya	
Department / HOD		

End of Software Requirements Specification